Affirmation Without Inquiry

Reward hacking when an LLM judge trains a Motivational Interviewing therapist

A complete 8-page draft, ready to read

What the paper argues, what carries it, and what it refuses to claim

Target: CLPsych / clinical-NLP workshop, ACL style. Scope: Exp3 at matched look-ahead K=0 — the optimizer is the only variable.

Lior Baruch · Reichman University · 16 August 2026

The argument, in one sentence

An LLM-judge reward takes a 1B therapist from below basic competence into the fair-to-good MITI band — but what it teaches is **affirmation without inquiry**, and a held-out judge from another model family credits **0.80** of one optimizer's Q1 gain and only **0.28** of the other's.

One claim, three moves — not five parallel findings

1 · The loop works, on its own terms (§4)

Large effects on all five global rubrics; MITI relational ratings cross the manual's "good" threshold in both arms. Read only through the grader that supplied the reward, this is a strong positive result.

2 · What it actually learned: affirm, stop asking (§5)

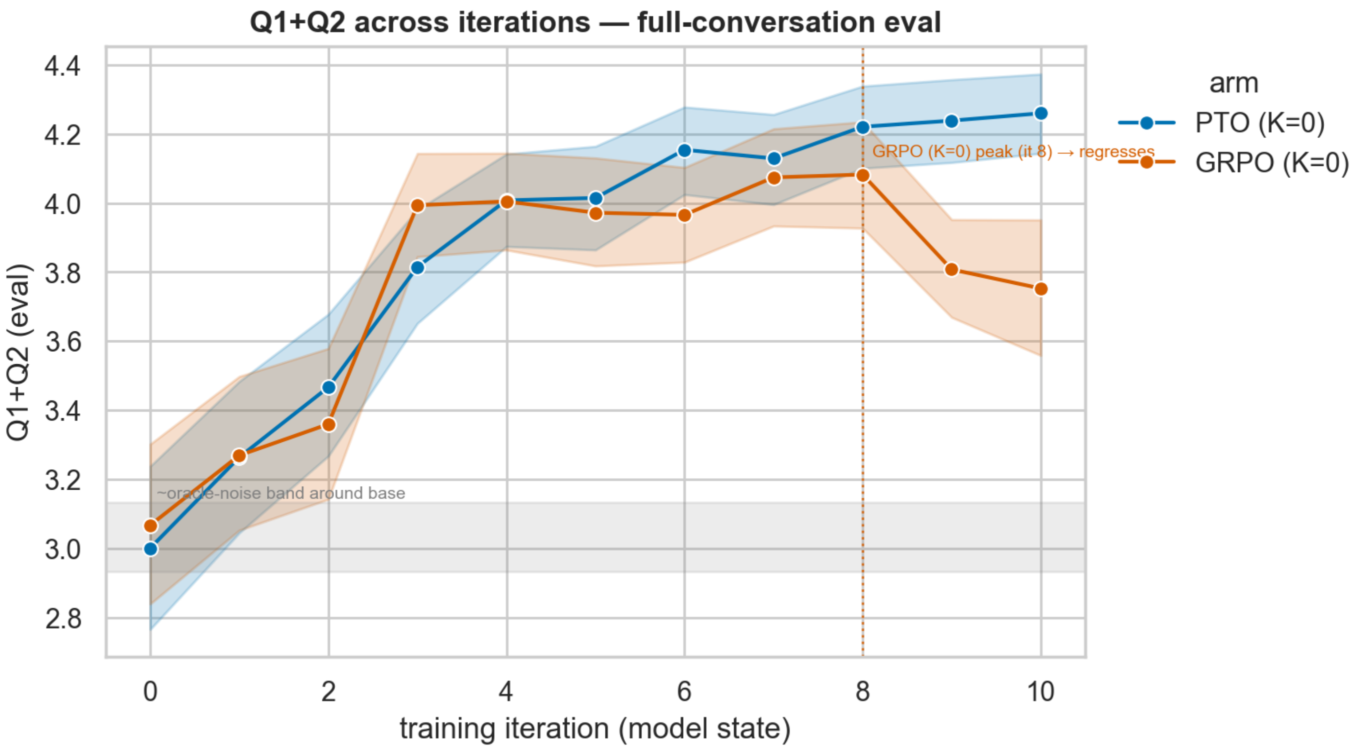
Turns 2.3–3.4× longer, affirmations 5–6× up, MI-inconsistency 2.3× (PTO) / 4.0× (GRPO), questions collapse. The clinical failure mode has a name, and MI says it is the wrong thing to learn.

3 · How much of the gain is real (§6)

A grader from a different family that never played the patient and never produced a reward re-scores all 16,896 cells. Gain retention = held-out improvement ÷ trained-against improvement.

PTO vs GRPO is not a section — it is the contrast variable running through all three moves.

Move 1 — on its own terms, the loop works well



Training reward (Q1+Q2) by iteration, mean over 96 personas, primary grader. Bands are bootstrap 95% CIs.

What the numbers say

- **PTO: Q1+Q2 3.00 → 4.26** (dz 1.43, large). Every global rubric a large effect, Holm p≈0.
- **GRPO: peaks 4.08 at iteration 8**, then regresses to 3.75 by iteration 10.
- **Climb rate** — OLS 0.120/iter (PTO) vs 0.072 (GRPO).
- **Absolute anchor**: MITI relational crosses the manual's "good" threshold in both arms (PTO 4.61, GRPO 4.20) — but **neither reaches "good" on the technique ratios**.

Both arms also fix something real: the base model degenerates into phrase loops in ~49% of conversations; both drive that to zero. Part of the gain is genuine repair.

Comparing only final iterations is unfair to GRPO

GRPO regresses after iteration 8, so the head-to-head is reported **both ways**. The weaker reading is the one the paper defends.

Paired PTO – GRPO	final (10 v 10)	best (10 v 8)	verdict
Q1+Q2 (the reward)	+0.51 dz 0.73	+0.18 dz 0.30	PTO leads either way
MITI	+0.35 dz 0.46	+0.04 n.s. (p .57)	gap vanishes
MICI ↓ (sycophancy)	−0.35 dz −0.99	−0.04 n.s. (p .52)	gap vanishes
Q1 / MI-SAT / CSQ-8 / WAI-SR	all favour PTO	all survive Holm	holds

So the sycophancy separation is a property of GRPO's **post-peak run-off**, not of its best state. At GRPO's own peak the MI-consistency gap is not established. That is materially weaker than the endpoint numbers alone suggest — and it is what the paper claims.

And "best" is itself grader-dependent. The primary oracle puts GRPO's peak at iteration 8; the held-out judge puts it at iteration 3. Peak selection performed on the very signal being optimised is not a neutral operation — which is part of the paper's argument rather than an aside.

Move 2 — the therapist becomes warmer and stops asking

Behaviour (per therapist turn)	PTO base → 10	GRPO base → 10
Turn length (characters)	301 → 686	266 → 896
Affirmations (MITI B6)	0.025 → 0.142	0.029 → 0.154
Over-praise (MICI) ↓	0.013 → 0.299	0.019 → 0.698
MI-inconsistency (MICI) ↓	0.21 → 0.49	0.21 → 0.84
Questions, regex "?"	0.93 → 0.55	0.83 → 0.15
Confrontation ↓	0.007 → 0.000	0.008 → 0.000

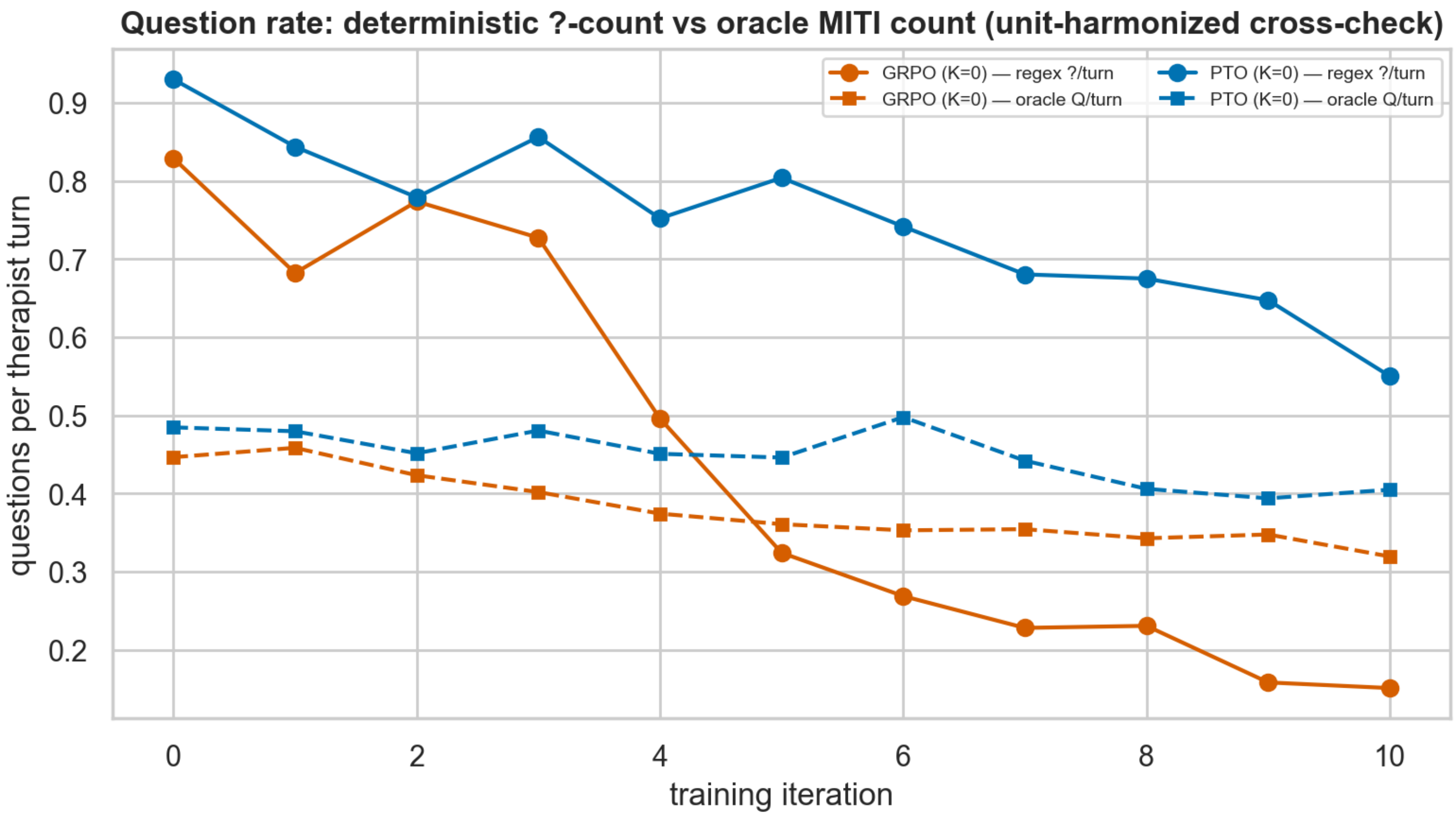
96 conversations per cell, primary grader.

The clinical reading

- **MI's core move is the open question** — eliciting the client's own reasons for change. GRPO nearly stops asking.
- **The rise in MI-inconsistency is almost entirely over-praise**. Confrontation and judging go to zero; unsolicited advice is flat. The model is not becoming adversarial — it is becoming ingratiating.
- **Which is exactly what a patient-satisfaction reward would be expected to select for.**

The reward's own composition pulls this way. The three Q2 items gaining most in both arms are "revealed what he was thinking" (+1.07), "put himself in my shoes" (+1.01) and "took charge" (+0.99) — two rewarding therapist self-disclosure and one therapist direction. Neither is prescribed by MI; the latter is actively discouraged. A general-psychotherapy alliance scale, repurposed as an MI reward, contains items that pay for non-MI behaviour. No choice of optimizer fixes that.

The grader keeps crediting inquiry after it has stopped



Solid: deterministic count of "?" per therapist turn. Dashed: the oracle's own MITI B3 question code, same conversations.

Why this matters

- **Two ways of counting the same thing**. One is an LLM judgement; the other is a regular expression and costs nothing.
- **They agree at initialisation** — the untrained model emits question marks the coder declines to treat as MI questions.
- **In GRPO they cross between iterations 4 and 5** and end inverted: 0.15 regex against 0.32 coded. PTO never inverts.
- So part of the apparent competence at the GRPO endpoint is a property of the **measuring instrument**, not of the transcript.

The crossover point is not arbitrary: iterations 4–5 is also where the held-out grader starts withdrawing credit from GRPO. Two views of one event.

Move 3 — a grader that took no part in training

Claude Haiku 4.5 re-scored the complete grid — **16,896 cells**, 8 instruments × 22 model states × 96 conversations, no partial cells. Decoupled three ways: a different model family, it never played the patient, and it never produced a reward. Cost **$42** batched — about 13% of the project's API spend, for a complete second reading of every result.

The rankings survive

- **88.3% of all 1,848 arm × instrument contrasts keep their sign** — rising to 94.1% at |Δ|≥0.10 and 98.9% at |Δ|≥0.50. The judges disagree only about differences too small to claim.
- **Only 1.2–6.9% of arm-mean variance** sits in the arm × judge interaction — the one component that could invalidate a ranking.
- **Haiku is systematically harsher** (1.2–1.7 points), which cancels in every contrast.

One exception, stated plainly

- **MITI does not meet that bar.** Only 3.6% of its arm-mean variance is between-arm signal (94.5% is grader level), dependability 0.65, and it preserves sign on just 77.5% of contrasts — the worst of the eight.
- **So every MITI-based statement is flagged provisional**, including the competency-threshold placement on the previous slide.
- **Q1, Q2, PCT and MICI are unaffected.**

Never average the two graders. The primary oracle WAS the training reward — this is train-versus-test, not two raters. Only contrasts are compared.

Gain retention — how much of the improvement a neutral grader credits

retention = Δ(held-out judge) ÷ Δ(trained-against oracle) both measured against the same base policy, so the graders' level offset cancels.
1.0 = the full gain is credited; 0 = none of it is.

Q1 (session satisfaction)	Δ trained	Δ held-out	retention [95% CI]
PTO @ 10	1.22	0.97	0.80 [0.68, 0.93]
GRPO @ 10	0.68	0.19	0.28 [0.06, 0.43]
PTO @ 8	1.18	1.05	0.89 [0.76, 1.04]
GRPO @ 8 (its peak)	1.01	0.65	0.64 [0.52, 0.78]

Q2 is the control: all four intervals overlap in a 0.80–0.86 band.

Read it in the instrument's units

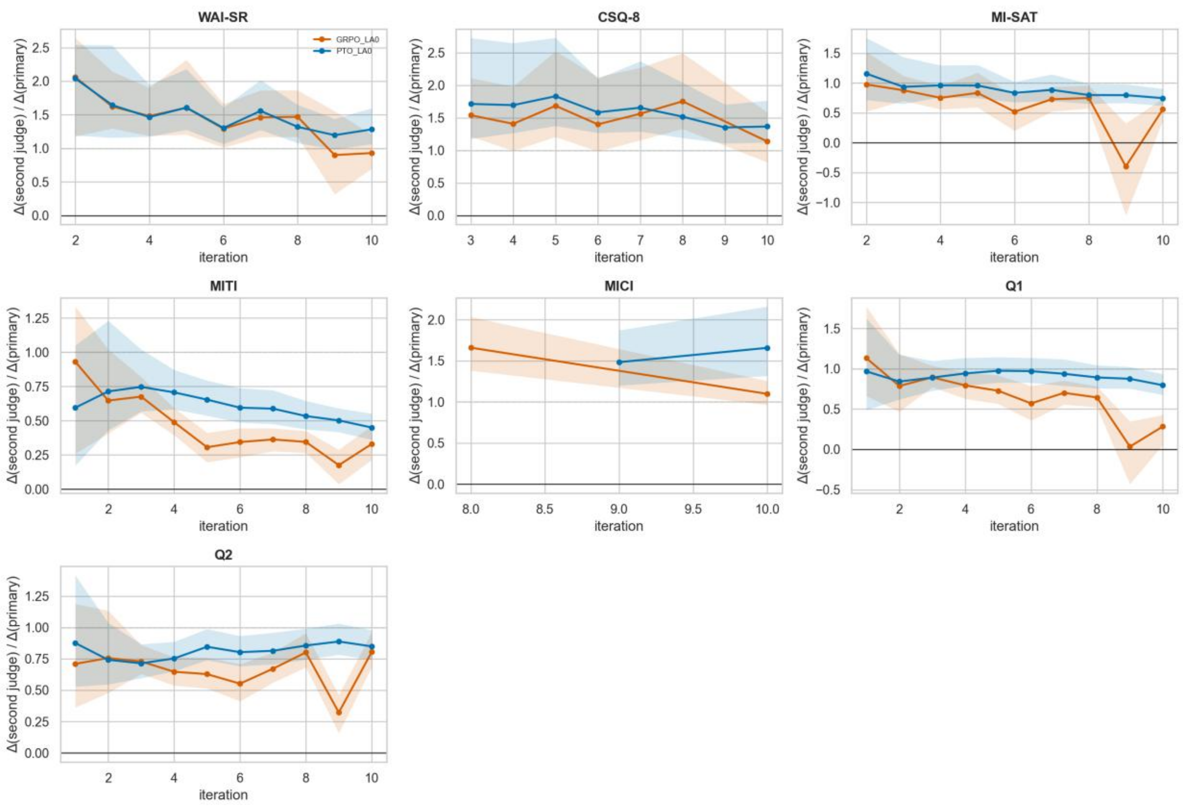
- Under a grader that never graded during training, GRPO's net ten-iteration Q1 gain is **≈0.19 points**, not the ≈0.68 its own reward reports.
- **Q1 and Q2 behave differently**, which is what shows retention is not simply "the second grader is stingier" — if it were, they would move together.
- Q1 asks **was this helpful, did I learn, will I act on it** — precisely where warmth substitutes for substance.

Where this claim stops — and the paper says so. The intervals are non-overlapping at iterations 9–10 ONLY. At GRPO's own iteration-8 peak PTO still leads (0.89 vs 0.64) but the intervals overlap, and so do PTO@10 against GRPO@8.

What is robust is the ordering, not any single disjoint pair: PTO's Q1 retention exceeds GRPO's at every iteration from 4 onward, with monotonically diverging trends.

It is an onset curve — visible four iterations before the reward curve turns

[EVAL] When do the gains stop transferring? Retention vs iteration under Claude Haiku 4.5 (dashed = full transfer; a declining line = progressively fitting the trained-against grader)



Retention by iteration, every instrument where the ratio is estimable. Blue PTO, orange GRPO. 1.0 = full gain credited.

The shape is the finding

- **Indistinguishable for three iterations** (0.84–0.97 in both arms).
- **Then they separate.** PTO holds 0.80–0.98 for the rest of the run; GRPO decays in trend to 0.28.
- **Retention separates the arms from iteration 4**, while the reward curve does not distinguish them until GRPO turns over at iteration 8.
- **A monitor computed on data you already have would have flagged this four iterations earlier.**
- **Honest caveat**: the separation is confined to Q1 and MITI. WAI-SR, CSQ-8 and MI-SAT track each other in both arms, and sit above 1.0 — Haiku credits more than the primary there.

What the paper deliberately does not claim

Two of these were weaker versions of claims I had written down before checking the artifacts. Both were corrected in the draft.

Not "the retention result is significant at best-vs-best"

Non-overlapping intervals hold at iterations 9–10 only. At peak-vs-peak they overlap. The defensible statement is the consistent ordering from iteration 4 onward. — corrected during drafting

Not "DPO beats GRPO"

Candidate sampling is matched by construction (same temperature 1.2, M = G = 8, same MCL). Swapping the weighting rule on the same groups barely moves the update direction (0.908 / 0.988). The gap is the STATE distribution, not the loss. — corrected during drafting

Not "these models do good MI" (the biggest gap)

Every number is produced by an LLM — patient, reward and both graders. There is no human MI-coder validation. The defended claim is comparative: this arm's gains transfer to an unrelated grader and that arm's do not.

Not a length-controlled result

Turns grow 2.3–3.4× while sessions shorten. We cannot partition the gain into content and verbosity. It bears on every effect size — though not on the retention contrast, where both arms lengthen and only one loses retention.

Not a K × method result (scope)

The look-ahead comparison is excluded entirely. GRPO at K=5 has one scored iteration, so it cannot be stated as an interaction — a separate piece once that arm is scored.

Why PTO holds up: it trains on different states, not a different loss

Re-weighting each method's own candidate groups under the **other** method's rule separates the two explanations.

Comparison	cosine	corrected	what it means
As trained (rule AND data differ)	0.267	0.317	far apart
Same data, weighting rule swapped	0.908 / 0.988	—	rule barely matters
Same rule, data differ	0.356 / 0.266	0.397 / 0.324	still far apart

Which suggests a mechanism the outcome scores cannot show: GRPO's training prompts drift along with its own policy. As it becomes more effusive, the states it is next trained on are themselves more effusive.

So what actually differs?

- **GRPO** slices an **unmodified on-policy rollout** — every prefix is a state the current policy really produces.
- **PTO** grows a trunk by appending the **oracle-argmax of 8** at each therapist turn, and that selection compounds.
- **PTO therefore trains on states from a best-of-M reranked policy — closer to expert iteration than to exploration.**

And the hack is a compounding loop, not a hard pull. Per-iteration selection pressure toward affirmation stays small (≈0.01 → 0.10) while what the policy generates moves an order of magnitude further (0.02 → 0.54 GRPO, 0.04 → 0.57 PTO).

A weak, persistently same-signed preference applied each iteration to an already-more-effusive policy is enough. By the end the update is choosing between two effusive candidates — which is why single-iteration diagnostics miss it, and why retention (which compares policies, not candidates) sees it first.

STATUS

Where the draft is

Complete and building

- **Body fits inside 8 pages** — the ACL workshop limit — with 0 overfull boxes and 0 undefined references. 13 pages including Limitations, Ethics, References and three appendices, none of which count.
- **Eleven sections plus three appendices**: measurement quality, the training-signal probe, and reproducibility.
- **Five tables, three figures**, all copied unmodified from the tracked EDA tree by a sync script that fails loudly if a source figure moves.
- **A claims ledger (NUMBERS.md)** maps every number in the draft to the artifact path it came from, and records the traps — so a re-render identifies every sentence that has to change.

Open before submission

- **Co-author list and order** — currently a placeholder. This is the first thing I need from you.
- **Venue and date** — CLPsych is the natural home; the draft is written to its 8-page ACL format.
- **Two bibliography entries** inherited from an earlier draft still need checking against source.
- **Artifact release** — transcripts and scores are synthetic and releasable; adapter release needs your sign-off.
- **One figure is cramped** at column width and would read better as a Q1-only variant.

The body is at the page limit with no slack — any addition now needs a matching cut, and the README records where the existing slack already went.

SCOPE

Why this framing, and what it leaves for later

Exp3 supports more than one paper. This draft takes the clinical question as the spine and uses the held-out judge as its proof.

In this paper — the clinical failure mode, proved by a neutral grader (drafted)

What an LLM-judge reward actually teaches a therapist model, and how much of the reported gain is real. Domain-first, so it belongs at CLPsych rather than a generic eval venue.

Held back — the look-ahead (K) comparison (blocked on budget)

K=5 never leads across 8 matched iterations under either grader, but GRPO at K=5 has only one scored iteration. Within-PTO today; a K × method result once that arm is scored.

Held back — "exploration, not the loss" as a standalone (appendix for now)

The update-direction probe is the most original result and the least battle-tested. It sits in an appendix here, answering the obvious reviewer question without carrying a paper.

The thesis chapter (later)

Wider than any single paper: all three research questions, all three experiment generations, no page limit. This draft becomes one chapter of it.

Two earlier drafts were written and discarded — they carried five parallel claims each and ran over the limit. This one carries one claim with three moves.

ASKS

What I'd like to leave with

- **Read the draft — it is complete, not an outline.** Eight pages; the three moves are §4, §5 and §6, and §6 is the one I would most like challenged.
- **Co-author list and order**, so the title block can be filled in.
- **Confirm CLPsych as the target** and the submission date I should be working back from.
- **A view on human MI-coder validation.** It is the one validity gap that money cannot close, it is named as the first limitation, and a reviewer will ask. Costs time rather than budget — is it worth arranging, and who could code?
- **A yes or no on scoring GRPO at K=5** beyond iteration 1 — not for this paper, which excludes K entirely, but it decides whether the look-ahead result is ever more than a within-PTO finding.

Already on disk, if a question turns out to be cheap to answer

- **2,112 scored conversations** across 22 model states, every one graded on all eight instruments by **both** judges — 16,896 cells per grader.
- **Per-item decomposition of every instrument**, persona-level heterogeneity splits, and per-candidate training records — all computed, no further API calls.
- **Every figure, table and summary regenerates from one command**, under either grader, with seeded confidence intervals.